Name: ____________________

PROBABILITY,
(BASIC PROBABILITY — SINGLE EVENTS)

Date: ______________

LO: I can calculate the probability of single events and place them on a probability scale.

What is probability?

Probability measures how likely an event is to happen, on a scale from **0 (impossible) to 1 (certain)**. It can be written as a fraction, decimal or percentage.

Formula: P(event) = number of favourable outcomes ÷ total number of possible outcomes.

Quick examples

	P(heads) = $\frac{1}{2}$	one favourable out of two outcomes
	P(6 on a die) = $\frac{1}{6}$	one out of six
	P(red from 3 red, 2 blue) = $\frac{3}{5}$	3 out of 5
	P(6 on a die) = $\frac{1}{3}$	there are 6 outcomes, not 3
	P(impossible) = 1	probability is never

Key idea

Probability = favourable outcomes divided by total outcomes. All probabilities lie between 0 and 1.

P(event) = favourable ÷ total

Always simplify the fraction where possible

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – fair die

A fair die is rolled. Find P(even number).

Even numbers: 2, 4, 6 = 3 outcomes

Total outcomes = 6

P(even) = $\frac{3}{6}$ = $\frac{1}{2}$

$\boxed{\text{P(even)} = \frac{1}{2}}$

Count the favourable outcomes and divide by the total.

Example 2 – bag of counters

A bag has 4 red, 3 blue, 2 green counters. Find P(blue).

Favourable (blue) = 3

Total = 4 + 3 + 2 = 9

P(blue) = $\frac{3}{9}$ = $\frac{1}{3}$

$\boxed{\text{P(blue)} = \frac{1}{3}}$

Add all counters for the total, then simplify.

Example 3 – complement

P(rain tomorrow) = 0.35. Find P(no rain).

P(not A) = 1 - P(A)

P(no rain) = 1 - 0.35

= 0.65

$\boxed{\text{P(no rain)} = 0.65}$

The probabilities of an event and its complement always sum to 1.

Example 4 – spinner

A spinner has sections: 2 red, 3 blue, 5 yellow. Find P(not yellow).

Total sections = 2 + 3 + 5 = 10

Not yellow = 2 + 3 = 5

P(not yellow) = $\frac{5}{10}$ = $\frac{1}{2}$

$\boxed{\text{P(not yellow)} = \frac{1}{2}}$

P(not yellow) = 1 - P(yellow) = 1 - $\frac{1}{2}$ = $\frac{1}{2}$.

YOUR TURN – PRACTICE (deliberate practice)

1. Fair dice
A fair six-sided die is rolled. Find each probability as a fraction.

a) P(3)

b) P(odd)

c) P(greater than 4)

d) P(less than 5)

2. Counters in a bag
A bag has 5 red, 3 blue, 2 white counters. Find each probability.

a) P(red)

b) P(white)

c) P(blue or white)

d) P(not red)

3. Complements
Find the probability of the complement.

a) P(A) = 0.3, find P(not A)

b) P(B) = $\frac{2}{5}$, find P(not B)

c) P(C) = 0.85, find P(not C)

d) P(D) = $\frac{7}{10}$, find P(not D)

4. Spinners
A spinner has 8 equal sections: 3 red, 2 green, 1 blue, 2 yellow.

a) P(red)

b) P(green or yellow)

c) P(not blue)

d) P(blue or red)

5. Mixed problems
Solve each probability problem.

a) Cards: P(heart) from a standard deck

b) Letters in MATHEMATICS: P(choosing M)

c) Numbers 1-20: P(multiple of 3)

d) P(rain) = 0.4, P(not rain)?

e) A bag has 6 red, 4 blue. P(red)?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Probability ranges from 0 to 1 ☐

Correct answer: __________

Reason: __________________

b) P(event) + P(not event) = 2 ☐

Correct answer: __________

Reason: __________________

c) P(heads on a fair coin) = 0.5 ☐

Correct answer: __________

Reason: __________________

d) If something is unlikely, P = 0 ☐

Correct answer: __________

Reason: __________________

e) P(event) = favourable outcomes ÷ total outcomes ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A bag contains **red, blue and green** counters. P(red) = 0.35 and P(blue) = 0.25. There are **40 counters** in total. How many green counters are in the bag?

Expression: ______________________________

Simplified: ______________________________

b) A spinner has sections labelled 1 to 8. The probability of landing on a prime number is $\frac{1}{2}$. Verify this is correct by listing all prime numbers from 1 to 8.

Expression: ______________________________

Simplified: ______________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!